

Experiment Name :

Experiment 1: Familiarization with Fundamental Logic Gates

Experiment 2: Universal Gates, Application of Boolean Algebra

Experiment 3: Parity bit checker and generator

Experiment -1 Objective

To get familiarized with fundamental logic gates and demonstrate the input-output relationship of 2-Input AND, OR and NOT gates by constructing their truth tables. Also we will get familiar with gates like NAND, NOR, XOR and XNOR

Required Components

1. IC - 7408 x 1
2. IC - 7432 x 1
3. IC - 7404 x 1
4. IC - 7400 x 1
5. IC - 7402 x 1
6. IC - 7486 x 1
7. IC - 7407 x 1

Experiment - 2 Objective

- (i) To investigate the rules of Boolean Algebra
- (ii) To gain experience working with practical circuits
- (iii) To simplify a complex functions using Boolean Algebra

Required Components

1. IC 7400 x 1

2. IC 7402 x 2

Experiment - 3 Objective

To design and implement an even parity generator and even parity checker using XOR gates (IC- 7486)

Required Components

1. IC - 7486 x 1

Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 1: Familiarization with Fundamental Logic Gates

Objective:

- To get familiarized with fundamental logic gates and demonstrate the input-output relationship of 2-input **AND** (IC – 7408), **OR** (IC – 7432) and **NOT/Inverter** (IC – 7404) gates by constructing their truth tables.
- To get familiar with other logic gates like **NAND** (IC – 7400), **NOR** (IC – 7402), **XOR** (IC – 7486) and **XNOR** (IC – 4077)

Required Components:

1. IC 7408 × 1
2. IC 7432 × 1
3. IC 7404 × 1
4. IC 7400 × 1
5. IC 7402 × 1
6. IC 7486 × 1
7. IC 4077 × 1

Procedure:

- For each of the ICs, place the IC correctly on the trainer board
- Remember to connect each IC's VCC pin to the "+5V" position of the DC Power Supply of the trainer board, and the GND or 0V pin to the "GND" position of the trainer board.
- Connect the inputs to the data switches and the output to any position on the LED display.
- Find out the outputs for all possible combinations of input states.
- Write down the input-output in tabular form.

Report:

Your report must have the following segments:

1. Name of the Experiment.
2. Objective.
3. Required Components.
4. Experimental Setup - Draw the pin diagrams of each IC.
5. Results - Truth Table of each IC.
6. Discussions.

Logic gate symbols with corresponding truth tables:

NOT



INPUT		OUTPUT
A		
0		1
1		0

AND



INPUT		OUTPUT
A	B	
0	0	0
1	0	0
0	1	0
1	1	1

OR



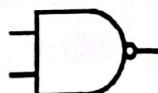
INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	1

XOR



INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	0

NAND



INPUT		OUTPUT
A	B	
0	0	1
1	0	1
0	1	1
1	1	0

NOR



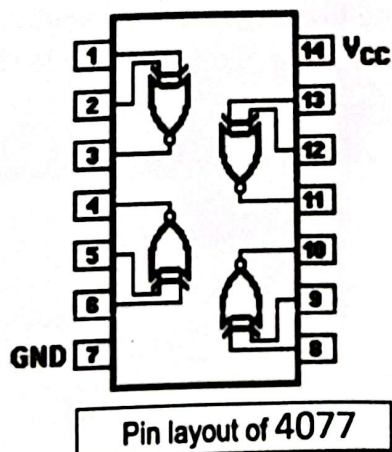
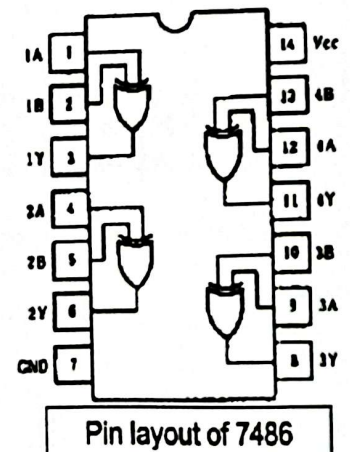
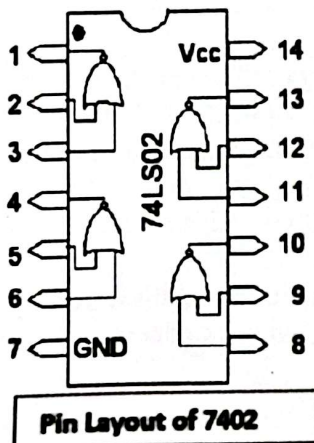
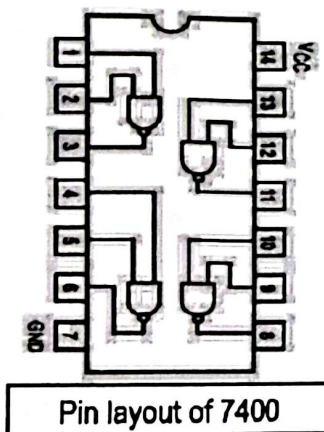
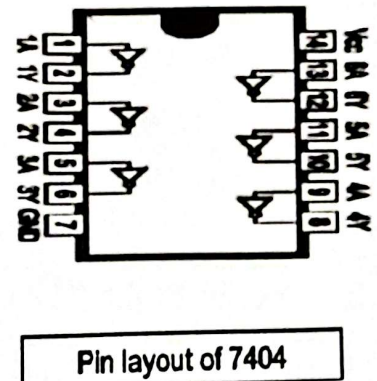
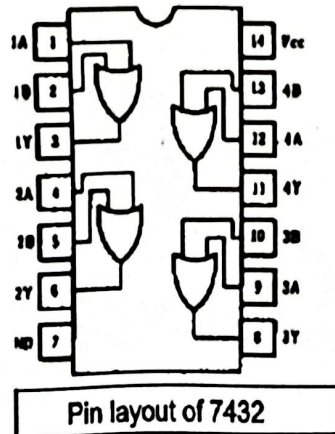
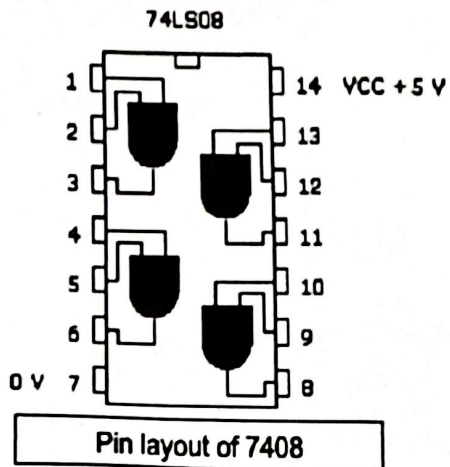
INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	0

XNOR



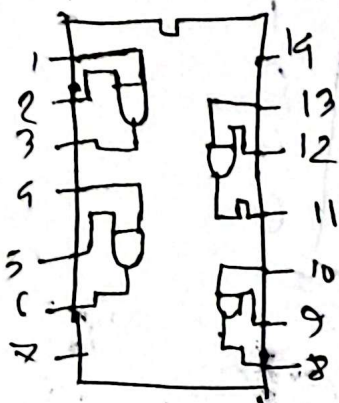
INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	1

Pin Diagrams of ICs:

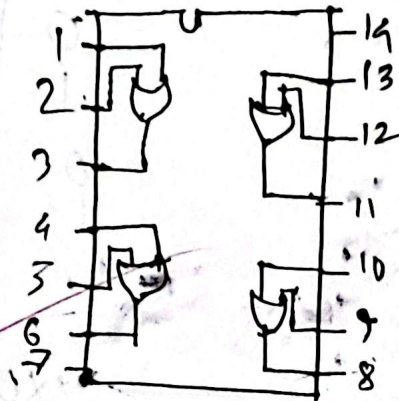


AND

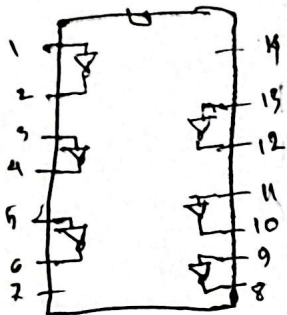
OR



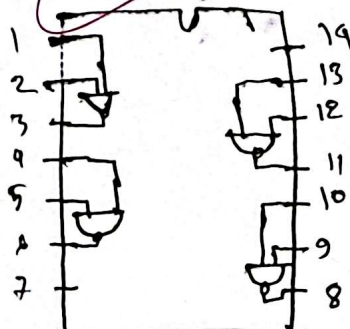
7408



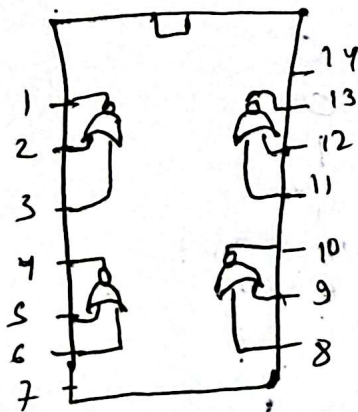
7432



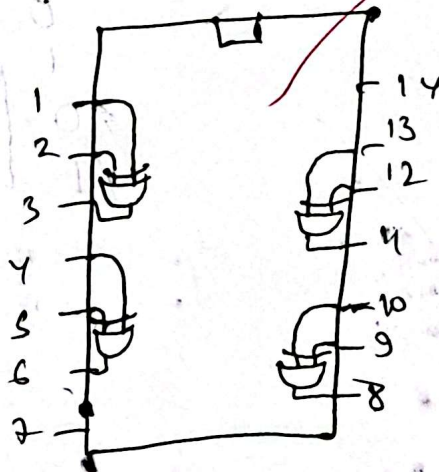
NOT 7409



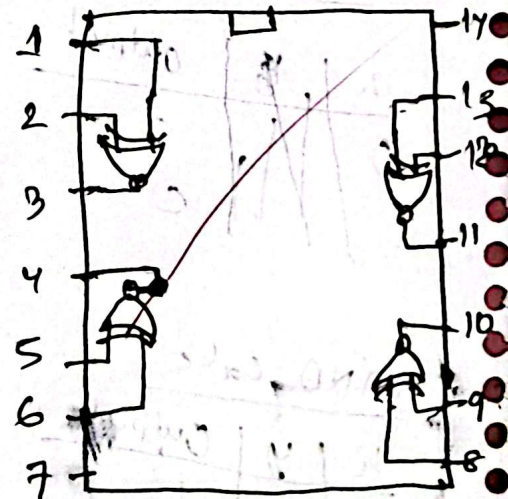
7400 (NAND)



NOR
7402



XOR
7486



XNOR
4077

AND Gate

x	y	output
0	0	0
1	0	0
0	1	0
1	1	1

OR Gate

x	y	output
0	0	0
1	0	1
0	1	1
1	1	1

Not Gate

x	output
0	1
1	0

NAND Gate

x	y	output
0	0	1
1	0	1
0	1	1
1	1	0

NOR Gate

x	y	output
0	0	1
1	0	0
0	1	0
1	1	0

XOR Gate

x	y	output
0	0	0
1	0	1
0	1	1
1	1	0

XNOR Gate

x	y	output
0	0	1
1	0	0
0	1	0
1	1	1

unlabeled

Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 2: Universal Gates, Applications of Boolean Algebra

Objective:

- To investigate the rules of Boolean algebra.
- To gain experience working with practical circuits.
- To simplify a complex function using Boolean algebra.

Required Components:

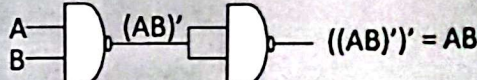
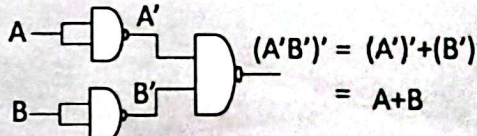
1. IC 7400 × 1
2. IC 7402 × 2

Boolean Postulates & Theorems:

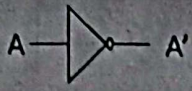
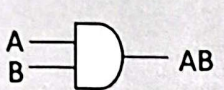
Postulate 2:	(a) $x + 0 = x$	(b) $x \cdot 1 = x$
Postulate 3: Commutative	(a) $x + y = y + x$	(b) $x \cdot y = y \cdot x$
Postulate 4: Distributive	(a) $x(y+z) = xy + xz$	(b) $x + yz = (x+y) \cdot (x+z)$
Postulate 5:	(a) $x + x' = 1$	(b) $x \cdot x' = 0$
Theorem 1:	(a) $x + x = x$	(b) $x \cdot x = x$
Theorem 2:	(a) $x + 1 = 1$	(b) $x \cdot 0 = 0$
Theorem 3: Involution	$(x')' = x$	
Theorem 4: Associative	(a) $x+(y+z) = (x+y) + z$	(b) $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
Theorem 5: DeMorgan	(a) $(x + y)' = x' \cdot y'$	(b) $(xy)' = x' + y'$
Theorem 6: Absorption	(a) $x + xy = x$	(b) $x \cdot (x + y) = x$

Diagrams:

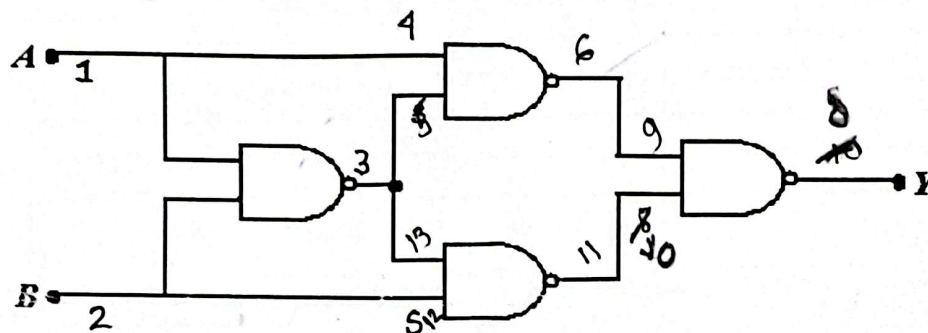
Building basic gates using Universal (NAND) Gate(s):

Not		
And		
Or		

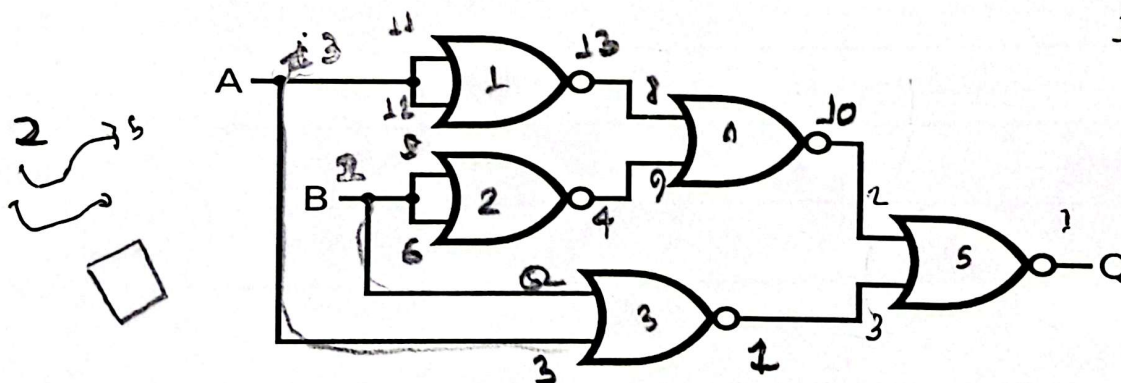
Building basic gates using Universal (NOR) Gate(s):

Not		$(A+A)' = A'$
Or		$(A+B)'$
And		$((A+B)')' = A+B$

Circuit Diagram - 1:



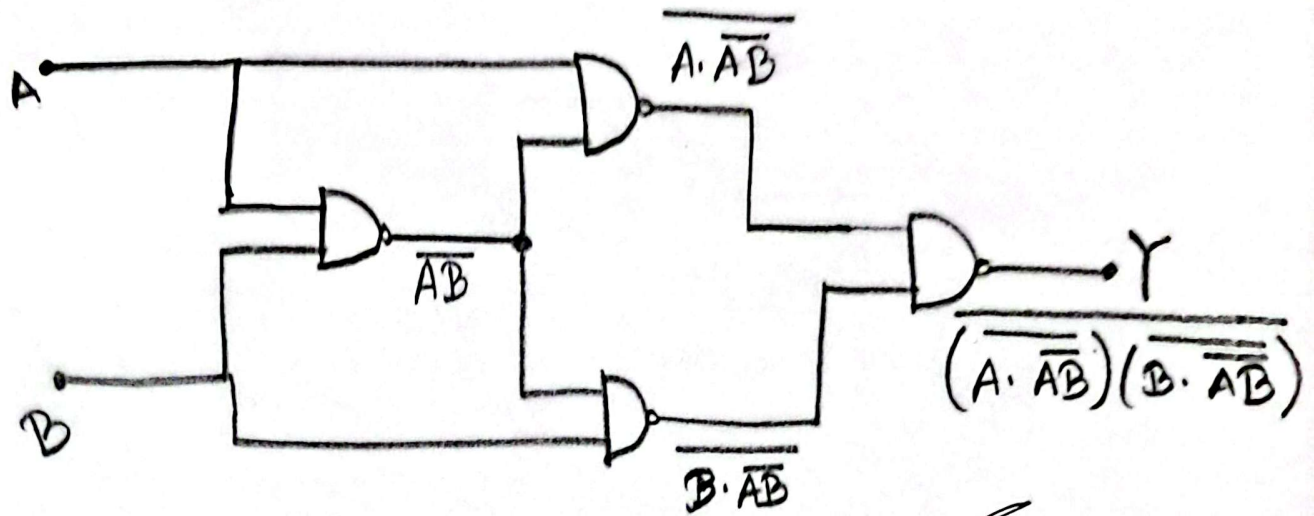
Circuit Diagram - 2:



Procedure:

- Construct the Circuit Diagram - 1 on the breadboard.
- Remember to connect each IC's VCC pin to the "+5V" position of the DC Power Supply of the trainer board, and the GND or 0V pin to the "GND" position of the trainer board.
- Connect the inputs to the Data switches and outputs to any position of the LED Display.
- Find out the outputs for all possible combinations of input states.
- Write down the input-output in tabular form.

Circuit Diagram - 1



~~10-25~~

Truth Table

A	B	B'	A'	AB'	BA'	AB' + BA'
0	0	1	1	0	0	0
0	1	0	1	0	1	1
1	0	1	0	1	0	1
1	1	0	0	0	0	0

Simplification

$$\overline{(A \cdot \overline{AB})(B \cdot \overline{AB})}$$

$$= \overline{A \cdot \overline{AB}} + \overline{B \cdot \overline{AB}}$$

$$= A \cdot \overline{AB} + B \cdot \overline{AB}$$

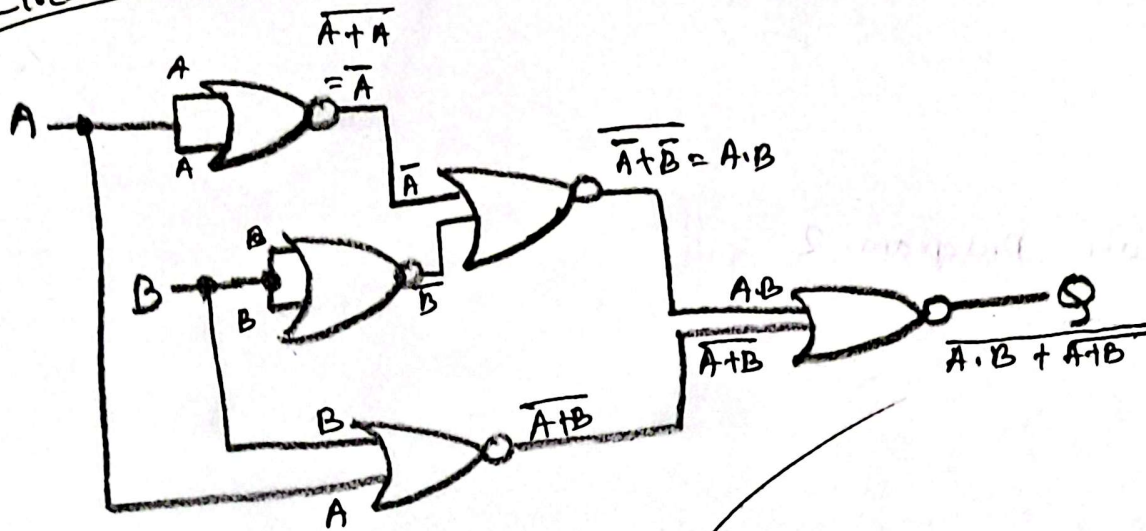
$$= A \cdot (\overline{A} + \overline{B}) + B \cdot (\overline{A} + \overline{B})$$

$$= A\overline{A} + \overline{B}A + B\overline{A} + B\overline{B}$$

$$= A\overline{B} + B\overline{A}$$

$$= A \oplus B$$

Circuit 2



Simplification

$$= A \cdot B + \bar{A} \cdot B + A \cdot \bar{B}$$

$$= A \cdot B + \bar{A} \cdot \bar{B}$$

$$= (A \oplus B) =$$

$$[(x+y)' = \bar{x} \cdot \bar{y}]$$

Truth Table

A	B	AB	$\bar{A}$	$\bar{B}$	$\bar{A} \cdot \bar{B}$	$A \cdot B + \bar{A} \cdot \bar{B} (A \oplus B)'$	$A \oplus B$
0	0	0	1	1	1	1	0
0	1	0	1	0	0	0	1
1	0	0	0	1	0	0	1
1	1	1	0	0	0	1	0

[Handwritten signature]

Simplification

$$= \overline{A \cdot B + (\bar{A} + \bar{B})}$$

$$= A \cdot B + (\bar{A} \cdot \bar{B})$$

$$= (A \cdot B)' \cdot (A' \cdot B')'$$

$$= (A' + B') (A'' \cdot B'')$$

$$= (A' + B') \cdot (A + B)$$

$$= A'A + A'B + B'A + B'B + A'B + A$$

$$= A'B + AB'$$

$$= A \oplus B$$

[F.O. = (p, q)]

A	B	$A \cdot B$	$\bar{A} \cdot \bar{B}$	$A \oplus B$	$A + B$	$A \cdot B + \bar{A} \cdot \bar{B}$	$\bar{A} \cdot \bar{B} + A \cdot B$	$\bar{A} \cdot \bar{B}$	$A \cdot B$	A	B
1	1	1	0	0	1	1	0	0	1	1	1
1	0	0	0	1	1	0	0	0	0	1	0
0	1	0	0	1	1	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1	0	0	0

Experiment # 3: Parity Bit Checker and Generator

Objective:

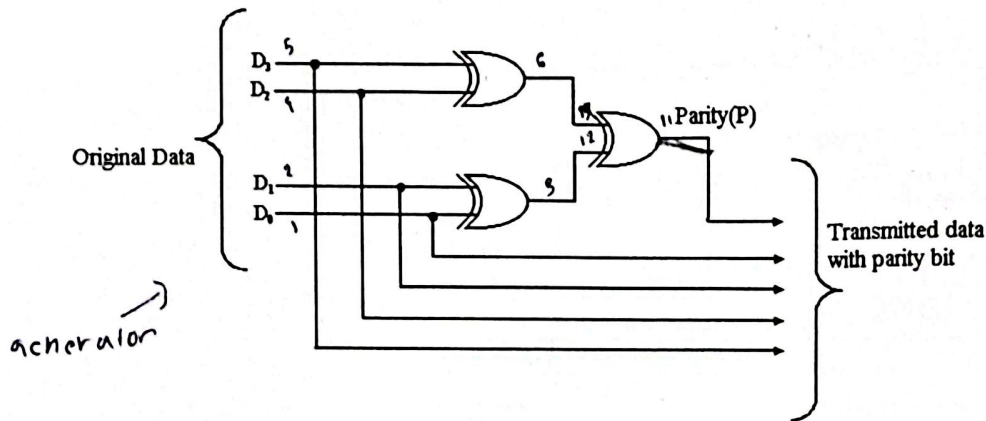
- To design and implement an even parity Generator and even parity checker using XOR gates. (IC-7486).

Required Components:

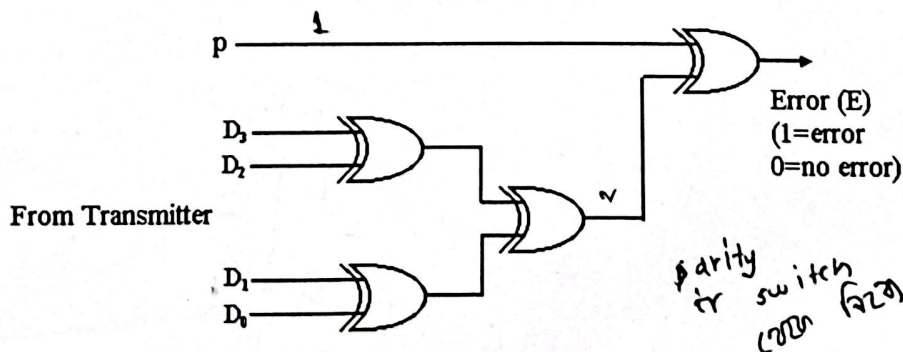
1. IC 7486 $\times$ 1

Diagrams:

Building an even parity generator using XOR Gates for 4-bit Data:



Building an even parity checker using XOR Gates for 4-bit Data:



Procedure:

- Construct the Circuits of both the parity bit generator and checker on the breadboard.
- Remember that each IC's pin 14 is connected to the "+5V" position of the DC Power Supply of AT-700, and pin 7 is connected to the "GND" position.
- Connect the inputs to the Data switches and outputs to any position of the LED Display.

Result:

Complete both of the following truth tables.

Truth Table for Parity Bit Generator:

	Data				Parity
	D ₃	D ₂	D ₁	D ₀	
a.	1	0	0	1	0
b.	0	0	0	1	1
c.	1	1	1	1	0
d.	0	0	0	0	0

Truth Table for Parity Bit Checker:

	Data					Error
	Parity	D ₃	D ₂	D ₁	D ₀	
a.	1	1	0	0	1	1
b.	0	0	0	0	1	1
c.	0	1	1	1	1	0
d.	1	0	0	0	0	0

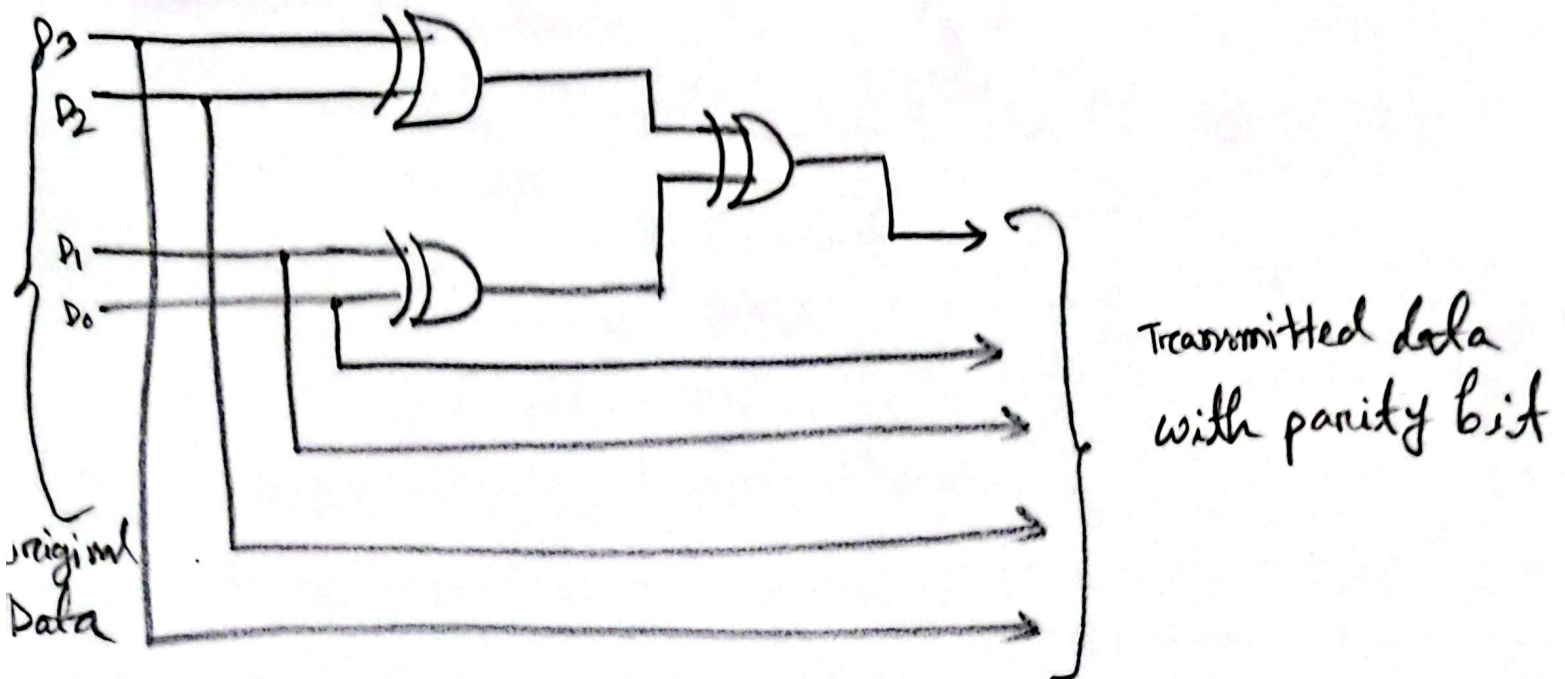
error

P	1	0	0	0	1	1
0					0	1
1	1	0	1	1		

0 11 10
0

Diagram:

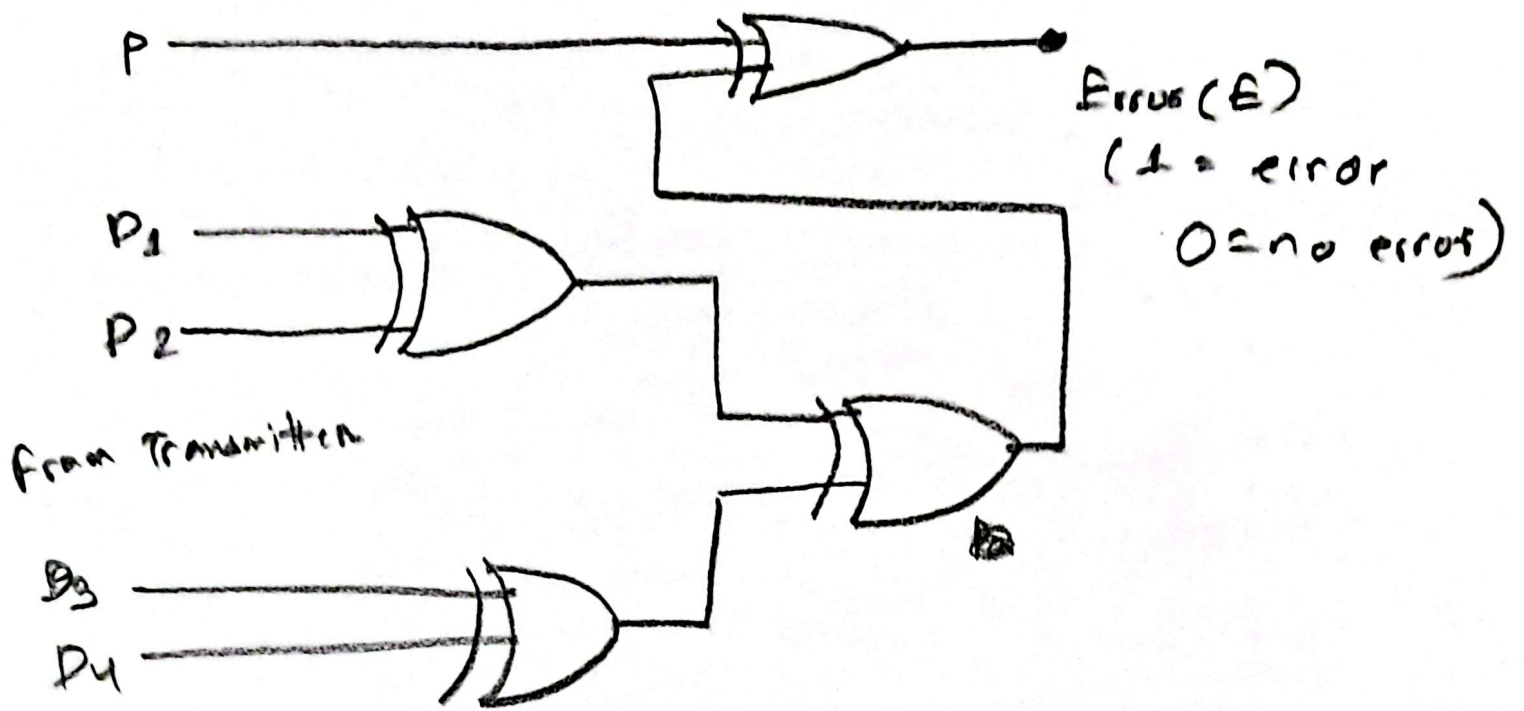
4-bit Data-even parity generator using XOR Gates.



Truth table for Parity Bit Generator:

	Data				Parity
a.	D ₃	D ₂	D ₁	D ₀	
a	1	0	0	1	0
b	0	0	0	1	1
c	1	1	1	1	0
d	0	0	0	0	0

~~5-11-25~~



Data					Error
Parity	P_1	P_2	P_3	P_4	
1	1	0	0	1	1
0	0	0	0	1	1
0	1	1	1	1	0
1	0	0	0	0	1

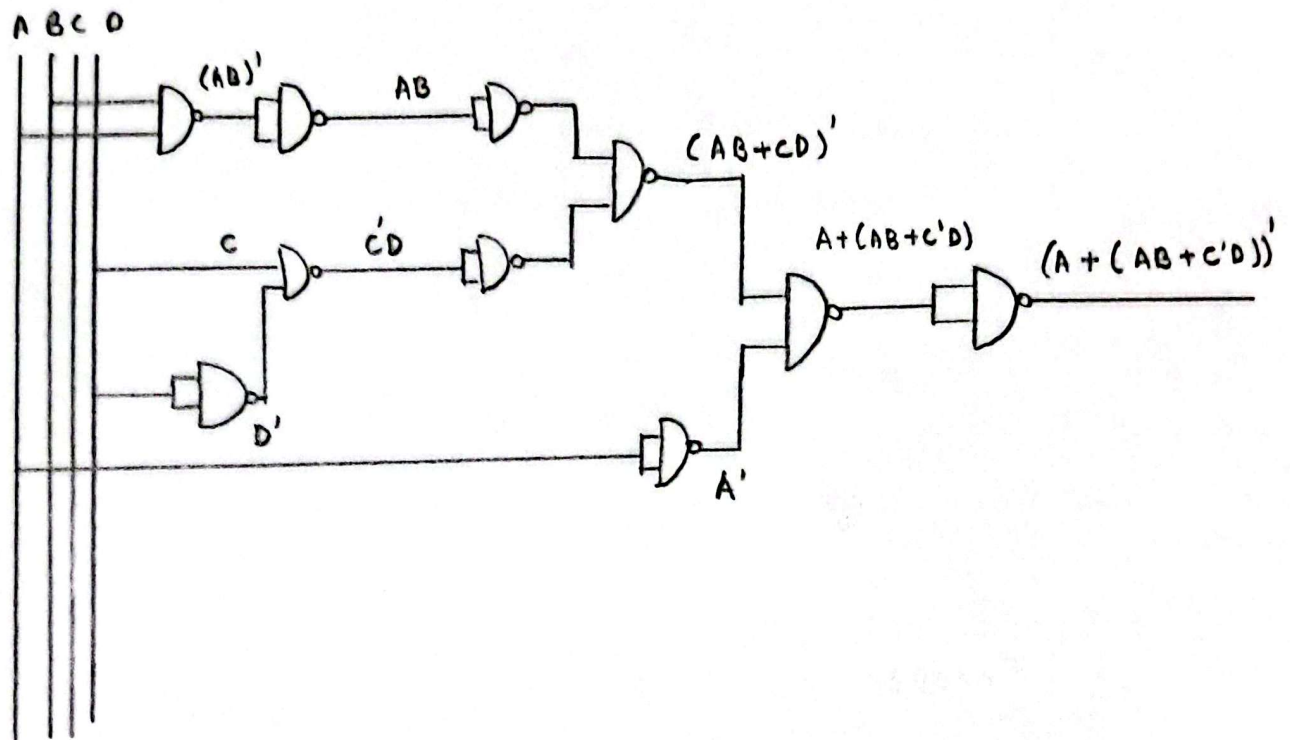
1

~~Alta~~
5-11-26

1
1
0
1

Answer to the Question No. 1

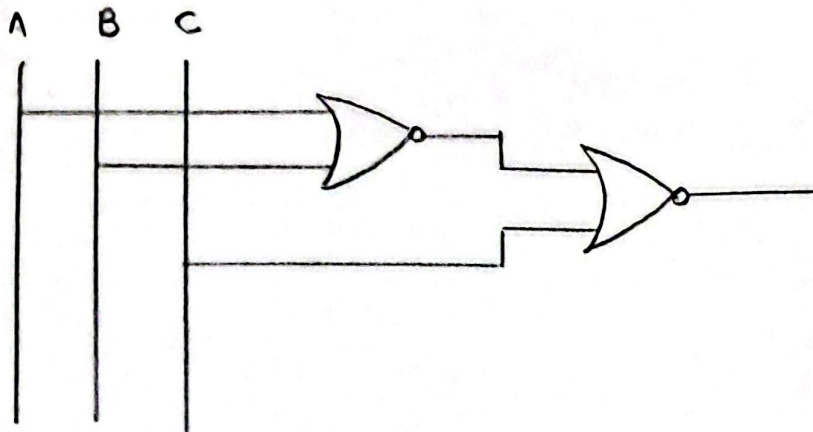
$$F = (A + (AB + C'D))'$$



Answer to the Question No. 2

NR-Parity Checker & Generator

Generator



Checker

